Leakage-Free Evaluation of Layer Choice in Protein Language Models for Mutation Effect Prediction

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Abstract

Motivation: Multi-layer embedding fusion is widely used with protein language models (PLMs) such as ESM-2, on the assumption that different layers carry complementary mutation-relevant information. This assumption has not been tested systematically for mutation effect prediction.

Results: We benchmarked ESM-2 and the structure-aware SaProt (650M parameters each) on 63 ProteinGym substitution assays using per-layer probing under four train–test splits, multi-scale pooling, a logit lens, and a position-lookup control. Under the standard random split all 33 layers performed alike ($\rho \approx 0.60$), but random position vectors reproduced this level and collapsed to 0.10–0.14 under leakage-free splits: random-split layer interchangeability is a property of the evaluation, not of the representations. Under leakage-free splits, performance rose with depth (ESM-2: $\rho = 0.23$ at layer 0, 0.46 at layer 32), and the best layer (layers 27–32) exceeded the final layer by at most $\Delta\rho = 0.019$ (90% upper bound 0.036). This deficit is small relative to the cross-dataset standard deviation of 0.17. Multi-scale pooling did not reliably improve on the final layer under leakage-free evaluation (1 of 96 comparisons significantly better, 55 worse), whereas fusion gains appeared under the random split. Real 3Di structure tokens improved SaProt slightly under the random split and substantially under leakage-free splits (+0.06 to +0.13). A validated logit lens peaked at the final layer and reproduced published ProteinGym zero-shot scores on single-substitution assays. Depth matters, and the final layer is a sufficient choice.

Availability and Implementation: Code and per-dataset results are available at https://github.com/schwarzeteeguo-creator/esm2-layer-invariance and archived at Zenodo (DOI: 10.5281/zenodo.22844290).

1 Introduction

Protein language models (PLMs) pre-trained via masked language modeling (MLM) on millions of protein sequences have become indispensable for mutation effect prediction [1, 2, 3]. Among these, ESM-2 [3] is one of the most widely adopted, with a 650M-parameter architecture comprising 33 transformer layers that produce 1280-dimensional per-residue hidden states. These embeddings have been successfully applied to diverse tasks including variant effect prediction [4, 5], protein design, and functional annotation [6, 7], establishing ESM-2 as a foundational tool in computational biology.

The internal organization of information across transformer layers has been extensively studied in natural language processing, where the "BERTology" literature has revealed that different layers specialize in distinct linguistic phenomena: lower layers encode surface-level syntax, middle layers capture semantic content, and final layers refine task-specific predictions [8, 9]. This hierarchical view has motivated analogous investigations in protein language models. A growing body of work posits that different transformer layers in PLMs similarly encode complementary biochemical information, and that combining representations from multiple layers should therefore improve downstream tasks. Kumar and Jha [11] demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, while early layers (6–14) have been shown to encode local physicochemical properties such as hydrophobicity and secondary structure propensity [12]. Alsamkary, Elshaffei et al. [13] proposed hierarchical pooling and pooled attention architectures for fusing representations across multiple PLMs and reported up to a 12% Spearman improvement for protein–protein interaction prediction. More broadly, the strategy of extracting and fusing embeddings from multiple transformer layers, or from multiple PLMs [26, 27], has become common practice in PLM-based pipelines [14], often under the implicit assumption that each source contributes non-redundant information.

These claims, however, rest on an assumption that has not been systematically tested: that different ESM-2 layers genuinely encode *complementary* information for mutation effect prediction, rather than merely offering different views of the same underlying signal. If layers are functionally similar under a given readout and evaluation protocol, then multi-scale fusion may add cost without improving prediction. If layer complementarity appears only under specific readouts or split protocols, then layer selection should be treated as a conditional design choice rather than a universal rule. Resolving this question carries practical consequences: the computational cost of extracting and storing embeddings from multiple layers is substantial for large-scale screening efforts, and the proliferation of ad-hoc layer-selection heuristics complicates reproducibility.

To resolve this question, we conducted a large-scale benchmark on ProteinGym [15], a comprehensive collection of deep mutational scanning (DMS) experiments spanning diverse protein families, assay types, and organismal contexts. We employed three independent experimental paradigms (Figure 1): (1) per-layer supervised probing via Ridge regression, (2) multi-scale embedding pooling with controlled feature sets, and (3) logit lens zero-shot scoring. In parallel, we quantified layer-wise representational similarity through cosine similarity and CKA analysis. Together, these approaches ask how much mutation-relevant information each layer carries, how it changes with depth, and how the answer depends on the evaluation protocol. We find that

the random-split layer profile is largely position memorization, whereas leakage-free evaluation shows that predictive information rises with depth, that the final layer is a sufficient choice, and that structure tokens carry a measurable benefit that the standard protocol largely hides.

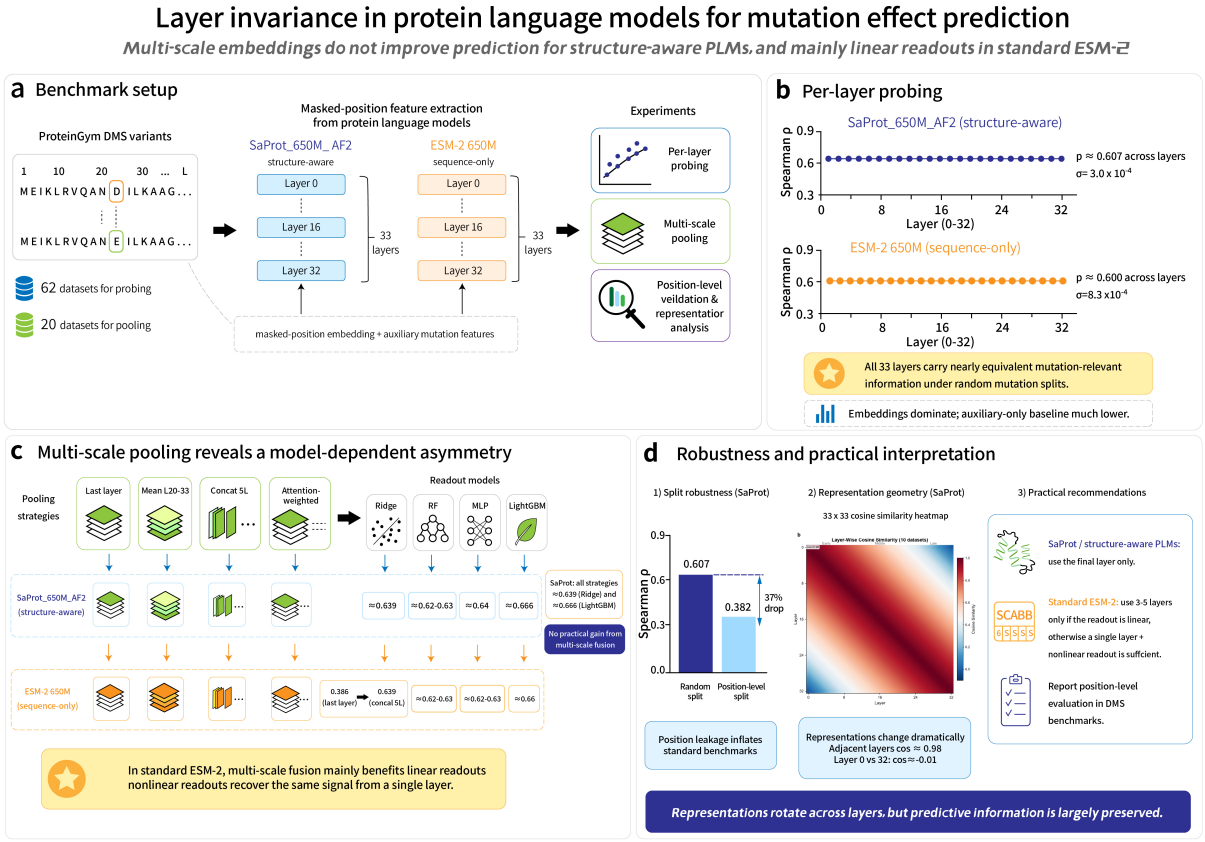


Figure 1: Experimental design overview. The three complementary experimental paradigms employed in this study: (**a**) Per-layer probing: Ridge regression on hidden states extracted from each of the 33 transformer layers under three feature arms (masked wild-type, unmasked wild-type, and mutant) and four train–test splits (random, modulo, contiguous, and position-grouped), quantifying the predictive information content per layer together with a random-position lookup control. (**b**) Multi-scale pooling: four embedding fusion strategies (Last Layer, Mean L20–33, Concat 5L, Attention-weighted, with attention weights trained within each cross-validation fold) tested across four readout models (Ridge, Random Forest, MLP, LightGBM) to assess whether combining layers improves prediction. (**c**) Logit lens: applying the final-layer LM head to intermediate hidden states for zero-shot mutation scoring, directly measuring how token predictions are refined across depth. (**d**) Layer-wise cosine similarity analysis: quantifying representational divergence independent of predictive performance. Together, these paradigms measure how mutation-relevant information is distributed across depth and whether combining layers adds to the final layer.

Alt text: Four-panel schematic of the study design: per-layer probing under three feature arms and four train–test splits with a position-lookup control (a), multi-scale pooling with four fusion strategies and four readouts (b), the logit lens (c), and layer-wise similarity analysis (d).

2 Methods

2.1 Models and Data

We used SaProt_650M_AF2 [16], a structure-aware variant of ESM-2 650M that extends the vocabulary to 446 structure-informed tokens (21 amino acids × 21 Foldseek 3Di states [17]), as our primary model. For validation, we also tested standard ESM-2 650M [3] in both the probing and logit lens experiments, enabling direct comparison between structure-aware and sequence-only representations. All experiments used ProteinGym's substitution DMS benchmark [15]. We used the 63-assay substitution snapshot that the SaProt repository distributes as LMDB files for supervised fitness prediction [16], which derives from an earlier ProteinGym release; the subset was fixed before any experiment and is identical across arms, models, and splits. The current release (v1.3) contains 217 substitution assays: 46 of our 63 assay identifiers are unchanged in v1.3, 13 correspond to assays that were renamed or reprocessed, and 4 (B3VI55_LIPST_Klesmith_2015 and three RL401_YEAST assays) are no longer included (per-assay status in Supplementary Table S1). Comparisons with published v1.3 results are therefore made on the 46 shared assays (Section 3.4). The assays contain experimentally measured fitness effects for single and multiple amino acid substitutions. All 63 datasets were used for the probing and logit lens experiments ($n = 63$ for the random, modulo, and position-grouped splits; six datasets whose mutated positions form a single contiguous block have undefined Spearman correlations under the contiguous split, which is therefore reported on the remaining 57). The pooling benchmark was run on a computationally tractable subset of 20 datasets because it crossed four pooling strategies with four readout models and multiple cross-validation folds. We therefore treat the pooling analysis as a controlled validation of the probing result rather than as a full ProteinGym-wide estimate. The subset comprises the first 20 datasets in alphabetical order, fixed identically for both models. Per-dataset mutation counts range from under 100 to over 200,000. For SaProt, we additionally prepared real Foldseek 3Di structure tokens by assigning each wild-type residue the 3Di state of its aligned residue in the AlphaFold Protein Structure Database entry for that sequence (structures available for 62 of 63 datasets; two aligned structures were truncated at the C terminus and the missing residues were padded with the Foldseek gap state; the influenza nucleoprotein dataset has no AFDB entry and was evaluated with structure-masked tokens). SaProt results are reported for both structure-masked and real-3Di inputs (per-dataset 3Di token lengths and gap counts in Supplementary Table S2).

2.2 Experiment 1: Per-Layer Probing

For each DMS dataset we extracted per-layer hidden states under three feature arms. The **masked_wt** arm replaces one mutated position at a time in the wild-type sequence with the mask token ("M#" for SaProt, `<mask>` for ESM-2) and reads the hidden state at that position (single-position masking, one forward pass per mutated position). The **unmasked_wt** arm reads the wild-type token at the mutated position without masking. The **mutant** arm feeds the full mutant sequence and reads the hidden state at the mutated position (averaged over the substituted positions for multi-site variants); it is the only arm whose representation differs between two mutations at the same position, and therefore the only one that can carry mutation-specific

information beyond position identity and the auxiliary substitution features. Multi-site variants are represented as one row per substituted position, and each row of the masked_wt arm receives the hidden state at its own substitution position. Both SaProt_650M_AF2 (structure-masked and real-3Di inputs) and standard ESM-2 650M were evaluated under identical conditions.

Each arm was evaluated under four train–test splits, all with five cross-validation folds (four for the single four-position dataset under the position-grouped split): the standard **random** split, which assigns whole variants to folds so that all rows of a variant share a fold; the **modulo** split, which assigns each residue position to fold (position − 1) mod 5 following the official ProteinGym protocol; the **contiguous** split, which divides the sequence into five contiguous segments and holds out one segment per fold; and a **position-grouped** split (GroupKFold over residue positions, so that all mutations at a given position share a fold). The latter three are leakage-free with respect to positions. In these position-based splits a row's fold is determined by its own substitution position, so rows of the same multi-site variant can fall into different folds and share a fitness label across folds. In the mutant arm, such rows also share the variant's averaged embedding. Under the position-based splits the mutant arm is therefore analyzed on single-substitution variants only (one row per variant, with the variant cap applied after filtering), and this is the primary mutant-arm analysis throughout. The wild-type arms, which share only the label, are reported on the full data, and all three arms are additionally reported on single substitutions (Section 3.5). For each layer $l \in \{0, \ldots, 32\}$ and feature configuration (combined: the hidden state concatenated with 42 auxiliary features, namely one-hot encodings of wild-type and mutant amino acids, the BLOSUM62 substitution score, and a normalized Grantham distance proxy; embedding_only; and auxiliary_only), we trained Ridge regression with the regularization strength selected per dataset by generalized cross-validation [23] (grid 10^{-2} to 10^{3}), standardization fit inside each training fold, and reported Spearman rank correlation ρ. Datasets were capped at 10,000 variants before row construction (11 of 63 datasets) and subsampled to 5,000 rows inside the pipeline (26 of 63 datasets reach this ceiling), with a fixed seed shared across arms, splits, and layers. The analysis comprises 213,393 rows from 203,056 variants; multi-site variants occur in 10 of the 63 datasets, where they make up most variants, and 49,131 analyzed rows (23.0%) come from multi-site variants (per-dataset counts in Supplementary Table S1). Because the 5,000-row subsample is drawn over rows, most of these rows appear without the other rows of their variant but carry the fitness of the full multi-mutant. The single dataset with fewer than five mutated positions (four positions) used four position-grouped folds.

To quantify how much probing performance is attributable to position identity alone, we added a **random-position control**: each mutation is represented by a fixed random vector indexed by its residue position (33 independent draws), probed with the identical Ridge protocol with and without auxiliary features. This control contains no sequence information; any performance it recovers measures position memorization.

2.3 Experiment 2: Multi-Scale Pooling

We compared four embedding strategies using fixed auxiliary features, evaluated on the **mutant** arm under the modulo split as the primary leakage-free protocol, and additionally under the

contiguous and random splits and on the masked_wt arm with the same pipeline: (1) **Last Layer**: the standard approach using only the final transformer layer; (2) **Mean L20–33**: mean pooling across layers 20–33, motivated by Kumar and Jha [11]; (3) **Concat 5L**: concatenation of layers 6, 14, 20, 26, and 32 (0-indexed, including the final layer), spanning the full representational depth; and (4) **Attention-Weighted**: softmax-weighted pooling across all 33 layers, with the weights trained within each cross-validation fold.

For each strategy, we evaluated Ridge regression (regularized by generalized cross-validation as above), Random Forest (50 estimators, max depth 10), MLP (two hidden layers of 256 units, ReLU), and LightGBM (100 estimators, max depth 7), using the same five cross-validation folds as the probing experiment for every dataset. Two of the 20 datasets (A4_HUMAN_Seuma_2021 and DLG4_HUMAN_Faure_2021) contain multi-site variants; because strategies are compared within the same arm and rows, we report a sensitivity analysis excluding these two datasets rather than restricting the benchmark to single substitutions. Standardization was fit inside training folds, and the auxiliary features were identical across strategies, isolating the effect of embedding choice.

2.4 Experiment 3: Logit Lens

We applied the model's final-layer LM head to the hidden states of each intermediate layer, producing pseudo-logits for zero-shot mutation scoring. All computations ran in fp32, and for every dataset the final-layer pseudo-logits were validated against the model's official logits pathway (maximum $|\Delta \log P| < 10^{-3}$; all 63 datasets passed for both models). For standard ESM-2, we used the conventional log-ratio, $\log P(\text{mutant}) - \log P(\text{wild-type})$. For SaProt with structure-masked input tokens (AA#, where # denotes masked structure), we marginalized over all 21 structure-token variants per amino acid; with real 3Di inputs, the wild-type structure token at each position was retained. We reported per-layer Spearman ρ between these zero-shot scores and experimental DMS fitness values across all 63 datasets. Scores follow the masked-marginal heuristic: each mutated position is masked in turn and each substitution is scored separately against its variant's fitness (cf. the tuned lens literature [25]).

2.5 Layer Similarity Analysis

We computed the 33×33 cosine similarity matrix of per-layer hidden states, averaged over 200 randomly selected residue positions in each of 10 datasets. Because raw cosine is not invariant to rotations of the coordinate system, we additionally computed linear CKA [24] and its debiased form based on the unbiased HSIC estimator [28] on the same hidden-state matrices. The standard CKA estimator is biased upward when the number of positions is small relative to the hidden dimension, so we report it at 200 positions per dataset, with a 50-position analysis for comparison. With fewer positions than hidden dimensions the debiased estimator is not estimable (its values exceed 1), and it is listed for completeness only (Supplementary Table S3). The unit of analysis for similarity-versus-distance statistics is the set of unique layer pairs (i, j) with $i < j$, diagonal excluded. A high CKA combined with a near-zero cosine indicates a change of basis that preserves linearly decodable information.

2.6 Statistical Evaluation

We used three complementary procedures: (1) **invariance score**, defined as the standard deviation of per-layer mean Spearman ρ (smaller indicates a flatter layer profile); (2) **equivalence testing** of the best layer against the final layer: for each model, arm, and leakage-free split we computed the per-dataset paired difference between the best layer and layer 32. The best layer is the argmax of the layer-mean curve computed on the same per-dataset results; this in-sample selection biases the comparison toward finding a difference, which makes equivalence verdicts conservative. The margin of $\Delta\rho = 0.01$ was fixed before the equivalence analysis was run, as the smallest difference that could plausibly change a downstream layer-choice decision; it is a practical convention rather than a task-derived threshold. Verdicts follow a single rule: *Equivalent* when the 90% bootstrap confidence interval of the mean paired difference (10,000 resamples, percentile method, fixed seed) lies within ± 0.01, *Difference* when it lies entirely outside the margin, and *Inconclusive* otherwise; two one-sided tests (TOST) p-values and Cohen's d_z are reported for information. The mutant arm enters these comparisons on single substitutions (Section 2.2). In 6 of the 27 model–arm–split combinations the best layer was the final layer itself, leaving 21 comparisons (Supplementary Table S4). Because several intervals end close to the margin, the verdicts obtained with the t-based interval and with TOST are reported alongside; and (3) **paired Wilcoxon signed-rank tests** for split contrasts (random versus leakage-free), for real-3Di versus structure-masked SaProt inputs, for each fusion strategy versus the last layer, and for readout contrasts, the latter two reported with 90% bootstrap confidence intervals of the mean paired difference.

3 Results

3.1 Representations Rotate Across Depth While Linearly Decodable Information Is Largely Retained

We first asked whether different transformer layers encode representations that are merely different views of the same underlying information, or whether they capture genuinely complementary features. To answer this, we computed the 33×33 layer-wise cosine similarity matrix of per-layer hidden states, averaged over 200 randomly selected residue positions in each of 10 datasets (Figure 2).

The matrix revealed a smooth representational trajectory across layers. Adjacent layers were highly similar (mean adjacent-layer cosine 0.979 for SaProt and 0.962 for ESM-2), and similarity decreased with layer distance (Pearson r over unique layer pairs: -0.50 for SaProt, -0.93 for ESM-2). The decay was smooth and monotonic, with no intermediate trough: for SaProt the mean cosine by distance fell from 0.97 ($d = 2$) through 0.81 ($d = 20$) to 0.50 ($d = 30$), collapsing to -0.008 between the most distant layers (Layer 0 vs. Layer 32). ESM-2 decayed more gently and remained positive throughout (0.93 at $d = 2$, 0.54 at $d = 32$). Crucially, near-zero raw cosine does not imply lost information: the linear CKA between SaProt layers 0 and 32 was 0.78 while their cosine was -0.008 (ESM-2: CKA 0.41, cosine 0.54; adjacent-layer CKA 0.97 for SaProt and 0.95 for ESM-2). With 50 positions per dataset the standard estimator, which is biased upward when positions are fewer than hidden dimensions, gave higher distant-layer

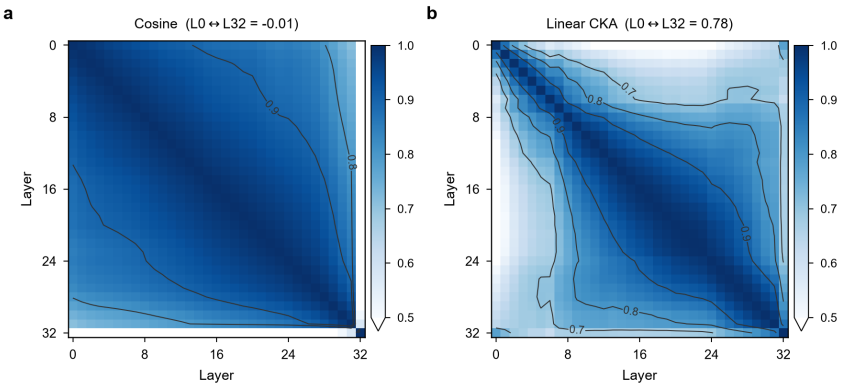


Figure 2: Layer-wise similarity matrices for SaProt (33×33). Averaged over 10 datasets and 200 random residue positions per dataset. (a) Cosine: adjacent layers are highly similar (cos = 0.98); mean similarity decays monotonically with distance and collapses to −0.01 only between the most distant layers (Layer 0 vs. 32). (b) Linear CKA over the same hidden states: the most distant layers retain CKA 0.78, indicating a rotation that preserves linearly decodable information rather than information loss. Both panels share a colour scale from 0.5 to 1 with contour lines at 0.7–0.9; cells below 0.5 (the final-layer row and column in panel a, and parts of the layer-0 and layer-1 rows and columns in panel b) are white.

Alt text: Two 33-by-33 heat maps of similarity between SaProt layers. Cosine similarity (a) stays high except between the final and first layers (about −0.01); linear CKA (b) remains moderately high throughout, 0.78 between layers 0 and 32.

values (SaProt 0.81, ESM-2 0.55); the debiased estimator is not estimable at either sample size (Supplementary Table S3). For SaProt, the depth-wise transformation is therefore well described as a rotation of the representation that retains much linearly decodable information (Figure 3c); for ESM-2, the most distant layers share less linear structure (CKA 0.41) despite a positive raw cosine. Similarity alone cannot show whether mutation-relevant information changes with depth; we therefore measured it directly with supervised probing, multi-scale pooling, and the logit lens.

3.2 Random-Split Layer Invariance Is Largely Position Memorization

We first asked whether the near-flat random-split layer profile reflects genuine representational redundancy. Under our pipeline the profile is flat: for the masked_wt arm, per-layer mean ρ spans 0.603–0.606 for both models with invariance scores $\sigma \approx 5\times10^{-4}$ to 1.2×10^{-3} (Table 2). A Ridge regressor given *random vectors indexed only by mutation position* plus the same auxiliary features recovered $\rho = 0.600$ under the random split (Table 1), matching the full masked-position PLM probing level. Under leakage-free splits the same control collapsed to $\rho = 0.138$ (modulo), 0.096 (contiguous, $n = 57$), and 0.135 (position-grouped), a paired drop of −0.46 (random versus modulo, Wilcoxon $p = 5.2 \times 10^{-12}$); the random vectors alone carried 0.557 under the random split and approximately zero elsewhere, and the auxiliary features alone fell from 0.339 to 0.19–0.23. Because every mutation at the same position shares one masked-position embedding, the random-split task is solvable by a position lookup table; the flat layer profile is therefore largely an artifact of position memorization rather than evidence that mutation information is distributed uniformly across depth.

Table 1: Random-position lookup control (63 datasets; contiguous $n = 57$). Mean Spearman ρ of Ridge regressors given no sequence information. The control reproduces the random-split PLM probing level and collapses under leakage-free splits.

Features	Random	Modulo	Contiguous	Position-grouped
Random position vector + aux	0.600	0.138	0.096	0.135
Random position vector only	0.557	−0.008	−0.056	−0.010
Auxiliary only	0.339	0.230	0.188	0.228

Table 2 and Figure 3a report the per-layer probing of all three arms, both models, and all four splits ($n = 63$; contiguous $n = 57$). Three patterns are stable across every leakage-free column. First, performance rises substantially with depth (Supplementary Tables S5–S7; Figure 5b): for ESM-2 masked_wt, ρ rises from 0.232 at layer 0 to 0.344 at layer 16 and 0.459 at layer 32 under the modulo split, and from 0.067 to 0.370 under the contiguous split; SaProt with real 3Di tokens rises from 0.274 to 0.460 (masked_wt, modulo). Second, the best layer sits in a narrow late band: layers 30–32 are best in 26 of 27 leakage-free model–arm–split combinations (layer 27 in the remaining one). Third, the best layer is at most marginally better than the final layer: the mean per-dataset difference was at most $\Delta\rho = 0.009$ for the wild-type arms and at most 0.019 for the mutant arm (ESM-2, position-grouped split; 90% upper bound 0.036). Of the 21 combinations in which the best layer was not the final layer, 7 were equivalent within the ± 0.01 margin, 14 were inconclusive, and none indicated a difference beyond the margin (Supplementary Table S4). Three of the seven equivalence verdicts have upper bounds within 0.0005 of the margin, and the t-based interval and TOST each classify one fewer comparison as equivalent, so these verdicts depend on the interval method. Deficits of this size are small compared with the cross-dataset standard deviation of about 0.17. Because these tests compare the best layer only with the final layer, they show that the final layer is a sufficient choice, not that layers are interchangeable. The invariance score rose from $\sigma \approx 10^{-3}$ (masked_wt, random) to $4\text{–}7 \times 10^{-2}$ under leakage-free evaluation, yet remained well below the cross-dataset standard deviation (≈ 0.17): performance varies far more *across* datasets than *across layers*, but under leakage-free evaluation the residual layer variation has a consistent direction, favoring the final layers. The mutant arm exceeded the wild-type arms under the random split (ESM-2: 0.681 versus 0.606) but not under leakage-free evaluation on single substitutions. There it was below both wild-type arms for ESM-2 (modulo, best layer: 0.443 versus 0.476 and 0.492) and level with them for SaProt with real 3Di tokens (0.492 versus 0.484 and 0.500; Supplementary Table S8). Under leakage-free evaluation, the mutant-sequence embedding therefore did not outperform the wild-type context embedding combined with the auxiliary substitution features.

3.3 Multi-Scale Pooling Does Not Reliably Improve on the Final Layer Under a Leakage-Free Protocol

We ran the pooling benchmark (mutant arm, modulo split, GCV-regularized Ridge, fold-internal standardization, attention weights trained per fold) on the 20-dataset design subset for each model (Table 3; Figure 3b). For SaProt (structure-masked inputs), all four strategies occupied a narrow band for every readout (Ridge spread across strategies 0.005; LightGBM 0.014). For ESM-

Table 2: Unified per-layer probing, combined features (mean Spearman ρ; best layer in parentheses, followed by the final-layer value when the best layer is not layer 32). $n = 63$; contiguous $n = 57$. Mutant-arm entries under the leakage-free splits (†) are computed on single-substitution variants (Section 2.2). Full 33-layer curves in Supplementary Tables S5–S7; per-dataset results in Supplementary Table S1.

Model	Arm	Random	Modulo	Contiguous	Pos.-grouped
ESM-2	masked_wt	0.606 (32)	0.459 (32)	0.370 (32)	0.450 (32)
ESM-2	unmasked_wt	0.606 (32)	0.470 (32)	0.388 (32)	0.461 (32)
ESM-2	mutant	0.681 (32)	0.443 (30) → 0.427†	0.353 (31) → 0.346†	0.444 (30) → 0.425†
SaProt (mask)	masked_wt	0.603 (31) → 0.603	0.391 (30) → 0.386	0.282 (31) → 0.274	0.385 (31) → 0.382
SaProt (mask)	unmasked_wt	0.604 (32)	0.412 (30) → 0.406	0.324 (31) → 0.320	0.403 (27) → 0.400
SaProt (mask)	mutant	0.652 (23) → 0.651	0.434 (30) → 0.428†	0.310 (30) → 0.307†	0.429 (30) → 0.423†
SaProt (3Di)	masked_wt	0.606 (32)	0.461 (31) → 0.460	0.394 (31) → 0.394	0.454 (31) → 0.453
SaProt (3Di)	unmasked_wt	0.606 (27) → 0.605	0.473 (31) → 0.466	0.415 (31) → 0.406	0.467 (31) → 0.459
SaProt (3Di)	mutant	0.684 (26) → 0.683	0.492 (30) → 0.486†	0.449 (31) → 0.439†	0.490 (31) → 0.481†

2, spreads were likewise small (Ridge 0.041; LightGBM 0.050), and the best single configuration was the last layer with LightGBM ($\rho = 0.508$), exceeding every multi-scale strategy under every readout. Per-fold-trained attention pooling was never the best strategy for either model. Paired per-dataset comparisons confirm this (Supplementary Table S9): no strategy–readout combination significantly improved on the last layer (largest mean difference +0.018, Concat 5L with Ridge, 90% CI −0.008 to +0.044, which does not exclude a gain of that size), and 14 of 24 combinations, all with nonlinear readouts, were significantly worse (90% CI below zero). Excluding the two datasets that contain multi-site variants left this unchanged (no combination significantly better; 17 of 24 significantly worse). Under the random split, by contrast, some fusion strategies did exceed the last layer: Concat 5L for ESM-2 with Ridge, Random Forest, and LightGBM (+0.009 to +0.012, 90% CIs excluding zero), and 10 of 12 strategy–readout combinations for SaProt (+0.010 to +0.064). The same benchmark on the masked_wt arm gave a similar picture (Supplementary Table S9). Under the modulo and contiguous splits no combination significantly improved on the last layer for ESM-2, and one did for SaProt (Concat 5L with Ridge, modulo split, +0.020, 90% CI +0.006 to +0.034). Across both arms, both models, and both position-based fold schemes, 1 of 96 strategy–readout comparisons favoured fusion and 55 favoured the last layer; one gain among 96 intervals is fewer than expected by chance at this interval level.

3.3.1 Readout Choice Matters More Than Layer Fusion for ESM-2 Under Leakage-Free Evaluation

Pooling comparisons in which feature construction, masking state, regularization, or split protocol differ between cells can manufacture apparent readout-specific layer dependences much larger than any effect seen when the pipeline is held fixed. With the pipeline held fixed, the largest single improvement for ESM-2 came from the readout rather than from fusion: on the last layer, LightGBM exceeded Ridge by +0.045 under the modulo split (paired 90% CI +0.021 to +0.068, Wilcoxon $p = 0.011$), whereas for SaProt the contrast was not significant (+0.016, CI −0.016 to +0.048, $p = 0.57$; Supplementary Table S10). The readout ranking is itself protocol-dependent: under the random split, last-layer LightGBM was below Ridge for both models (ESM-2 −0.035,

Table 3: Multi-scale pooling under the unified leakage-free pipeline (mutant arm, modulo split, combined features; 20 datasets per model; mean $\rho\pm$ sd). Spreads across strategies: ESM-2 Ridge 0.041, RF 0.057, MLP 0.045, LightGBM 0.050; SaProt Ridge 0.005, RF 0.010, MLP 0.039, LightGBM 0.014. Two of the 20 datasets contain multi-site variants; excluding them does not change the paired comparisons (Supplementary Table S9).

ESM-2 650M	Ridge	RF	MLP	LightGBM
Last Layer (baseline)	0.463 ± 0.174	0.476 ± 0.173	0.480 ± 0.198	$\mathbf{0.508} \pm 0.189$
Mean L20–33	0.473 ± 0.180	0.420 ± 0.165	0.436 ± 0.179	0.457 ± 0.174
Concat 5L	0.481 ± 0.183	0.453 ± 0.181	0.435 ± 0.170	0.495 ± 0.190
Attention-weighted	0.439 ± 0.201	0.419 ± 0.209	0.440 ± 0.209	0.463 ± 0.208
SaProt_650M_AF2	Ridge	RF	MLP	LightGBM
Last Layer (baseline)	0.432 ± 0.163	0.415 ± 0.174	0.445 ± 0.168	$\mathbf{0.448} \pm 0.178$
Mean L20–33	0.433 ± 0.166	0.415 ± 0.166	0.429 ± 0.166	0.439 ± 0.182
Concat 5L	0.434 ± 0.185	0.407 ± 0.173	0.406 ± 0.176	0.436 ± 0.183
Attention-weighted	0.429 ± 0.164	0.405 ± 0.166	0.426 ± 0.164	0.433 ± 0.177

SaProt -0.065; 90% CIs excluding zero).

3.4 Logit Lens Confirms Late-Layer Information Accumulation

The lens (fp32, per-dataset validation against the official logits pathway, $n = 63$) produced monotonically improving zero-shot scores with depth for both models (Figure 4). ESM-2 rose from $\rho = 0.031$ (layer 0) to $\rho = 0.443$ (layer 32), with only shallow fluctuations in early layers; SaProt with real 3Di tokens rose from -0.077 (layer 0) through approximately zero around layer 20 to $\rho = 0.437$ (layer 32). Both curves attain their maximum at the final layer, in agreement with the leakage-free probing: mutation-relevant information accumulates toward the final layers. Zero-shot absolute levels remain below supervised probing (0.44 versus up to 0.49), so the lens is best read as a consistency check on layer-wise trends rather than a standalone benchmark.

To anchor these values, we recomputed published ProteinGym v1.3 per-assay results on the 46 assays shared with our snapshot (Supplementary Table S11). The published zero-shot Spearman values on these assays are 0.472 for ESM-2 650M and 0.481 for SaProt 650M; our final-layer lens gives 0.438 and 0.440, and our zero-shot reproduction of SaProt with the authors' pipeline gives 0.484 (per-assay correlation with the published SaProt values $r = 0.98$). On the 38 shared assays that contain only single substitutions, the lens reproduces the published values (ESM-2 0.475 versus 0.476, per-assay $r = 1.00$; SaProt 0.473 versus 0.477, $r = 0.97$); the remaining gap arises in 8 multi-site assays, where the lens scores each substitution of a variant separately rather than summing over the variant. For supervised anchoring, the v1.3 release provides out-of-fold predictions of published supervised models under its official fold schemes. On the same 46 assays, Kermut reaches 0.605 and ProteinNPT 0.566 under the modulo split (0.561 and 0.525 under the contiguous split), above our leakage-free probing on the same assays (0.24–0.47 across models and arms), whereas the published one-hot baseline without augmentation falls from 0.570 under the random split to -0.003 under the modulo split, an external counterpart of our position-lookup control. On these assays, our final-layer mutant arm on single substitutions reaches 0.407 (ESM-2) and 0.469 (SaProt with real 3Di) under the modulo split (Supplementary Table S11).

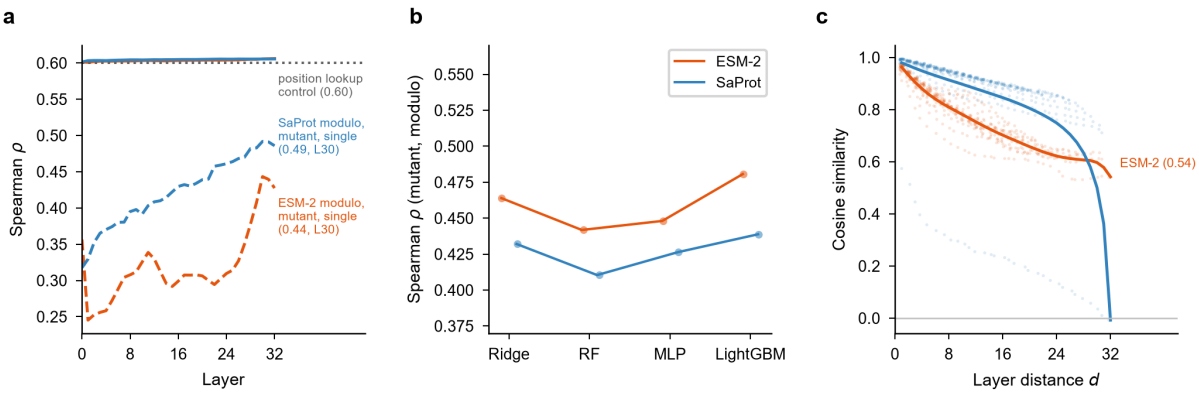


Figure 3: Per-layer probing, multi-scale pooling, and layer distance under the unified protocol. (**a**) 33-layer probing (combined features): under the random split the masked-arm profile is flat for both models ($\sigma \approx 10^{-3}$) and is reproduced by the random-position lookup control (dotted, 0.600); under leakage-free splits (modulo, mutant arm on single substitutions) performance is lower and the best layers are late (layer 30 for both models). (**b**) Pooling (mutant arm, modulo split, 20 datasets per model; four strategies as dots, mean connected): ESM-2 spread across strategies 0.041 (Ridge) to 0.057 (RF); SaProt spread 0.005 (Ridge) to 0.039 (MLP); last-layer LightGBM is the best single configuration; per-fold-trained attention pooling is never best. (**c**) Cosine similarity versus layer distance (unique layer pairs): SaProt decays monotonically and reaches −0.01 only at the largest distances, while ESM-2 remains positive; linear CKA between the most distant SaProt layers remains 0.78 (Figure 2), indicating rotation with preserved linear information rather than loss (200 positions per dataset).

Alt text: Three panels. (a) Under the random split, per-layer probing is flat at about 0.60 and matches the position-lookup control; under the modulo split, single-substitution mutant-arm curves are lower and peak at layer 30. (b) Pooling results by readout for both models. (c) Cosine similarity declines with layer distance.

3.5 Boundary Conditions

The results above show that what a benchmark reports about layer choice depends on how it is evaluated. We now examine the two factors that moderate every conclusion: evaluation protocol and input representation.

3.5.1 Leakage-Free Splits: What Is Lost and What Remains

The control and main-table results above already localize the leakage: the random-position lookup table reproduces the random-split level (0.600 versus 0.603–0.606 for full probing) and loses it entirely under leakage-free splits (0.10–0.14), while the random vectors alone carry 0.557 under the random split and approximately zero elsewhere (Tables 1 and 2; Figure 5). Among the leakage-free splits, modulo and position-grouped give similar levels (e.g., SaProt real-3Di unmasked_wt arm: 0.473 and 0.467) while contiguous is uniformly hardest (0.415), consistent with sequence-contiguous segments correlating with local structural context and mutational scope. Random mutation splitting, the de facto standard in DMS benchmarking, therefore inflates final-layer performance estimates by 0.14–0.26 ρ relative to the modulo and position-grouped splits and by 0.20–0.34 relative to the contiguous split and masks both layer differences and structure-token contributions. ProteinGym already provides the modulo and contiguous schemes used here [15]; results under the random split alone overstate performance and con-

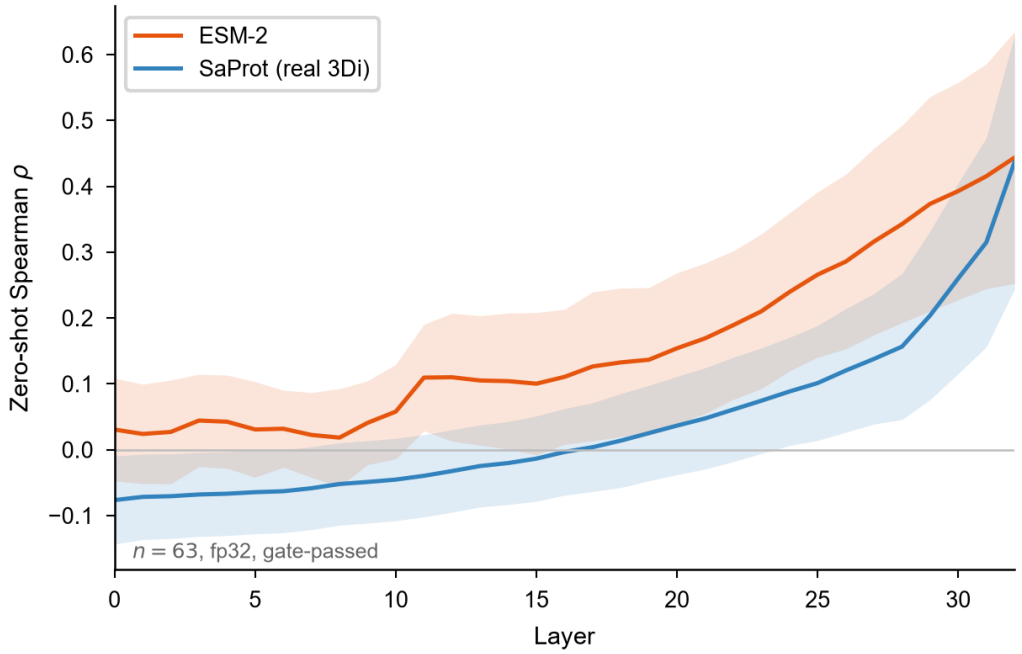


Figure 4: Logit lens: zero-shot mutation scoring by layer ($n = 63$). Both models improve with depth and attain their maximum at the final layer: ESM-2 (orange) rises from $\rho = 0.03$ (layer 0) to $\rho = 0.443$ (layer 32); SaProt with real 3Di tokens (blue) rises from negative early-layer values to $\rho = 0.437$ (layer 32). Shaded bands indicate ± 1 standard deviation across datasets. All computations in fp32 and validated against the official logits pathway.

Alt text: Line plot of zero-shot logit-lens Spearman correlation against layer for ESM-2 and SaProt; both curves rise with depth and peak at the final layer at about 0.44.

ceal the depth trend, a situation analogous to random versus temporal cross-validation for time series [18].

3.5.2 Multi-Site Variants and Single-Substitution Analysis

Because the position-based splits assign each row by its own substitution position, rows of the same multi-site variant can share a fitness label across folds. In the mutant arm they also share the variant's averaged embedding. Under the modulo split, 15,624 of the 213,393 analyzed rows (7.3%) are test rows whose variant also has a row in the training folds (15,960 under the position-grouped split and 6,158 under the contiguous split). Re-running all three arms on single-substitution variants only (Supplementary Table S8) shows that this inflated the full-data mutant arm (ESM-2, modulo, best layer: 0.483 on the full data versus 0.443 on single substitutions; SaProt with real 3Di: 0.531 versus 0.492). It did not inflate the wild-type arms (ESM-2 masked_wt: 0.459 versus 0.476), so only the mutant arm is analyzed on single substitutions under these splits. On single substitutions the depth trend and the late optimum were unchanged for all arms, with the best layer in layers 27–32 in every leakage-free column. The best layer exceeded the final layer by at most 0.019 (point estimates; 90% upper bounds up to 0.036; 3 of 23 comparisons equivalent, 20 inconclusive, none beyond the margin). Two datasets retain fewer than 50 single-substitution variants; excluding them leaves these conclusions unchanged (largest point estimate 0.014, 90% upper bound 0.025).

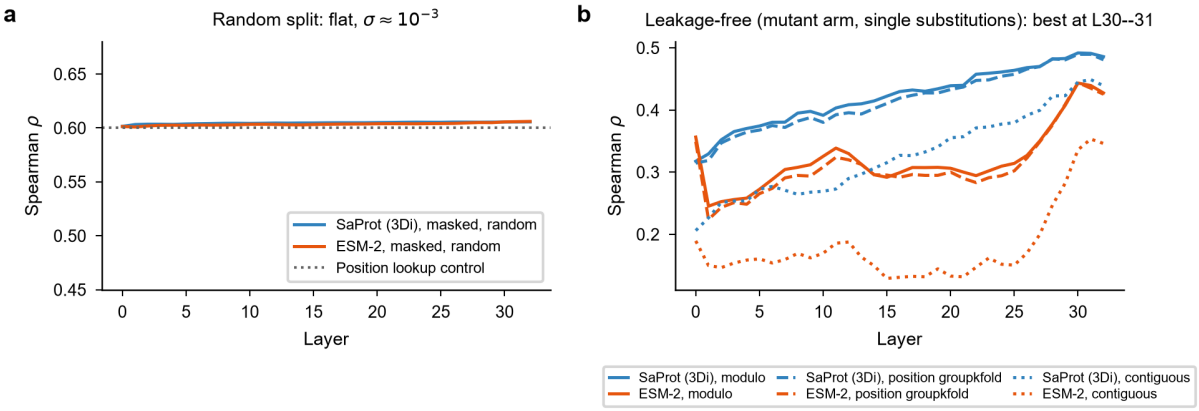


Figure 5: Random versus leakage-free split probing (ESM-2 and SaProt, 63 datasets). (**a**) Random mutation split: flat per-layer profile ($\sigma \approx 10^{-3}$), reproduced by the random-position lookup control ($\rho = 0.600$). (**b**) Leakage-free splits (modulo, position-grouped, contiguous; mutant arm on single substitutions): performance drops to 0.35–0.49, the invariance score rises to 4–7 $\times 10^{-2}$, and the best layers shift consistently to layers 30–31.

Alt text: Two panels of per-layer probing for ESM-2 and SaProt. (a) Under the random split, curves are flat at about 0.60 and match the position-lookup control. (b) Under leakage-free splits, single-substitution mutant-arm curves are lower (0.35 to 0.49) and peak at layers 30 to 31.

Together, these boundary conditions refine the central finding: apparent layer interchangeability under random splitting is dominated by position memorization, leakage-free evaluation reveals a consistent depth trend in which the final layer is sufficient, and the benefit of structure tokens depends strongly on the evaluation protocol.

3.6 Structure Tokens Help Most Under Leakage-Free Evaluation

Because SaProt can be run with real 3Di tokens or with structure-masked tokens, the unified protocol allows a paired test, dataset by dataset and layer by layer, of what structure tokens actually contribute (Table 4). Under the random split the differences were small (paired differences at layer 32: +0.003 masked_wt, +0.002 unmasked_wt, +0.032 mutant; Wilcoxon $p = 0.022$, 0.91, and 2.5×10^{-6}, so the effect is small but clearly non-zero for the mutant arm). Under every leakage-free split the real-3Di inputs were consistently superior, with mean paired advantages of +0.057 to +0.132 across arms and splits, all with Wilcoxon $p \leq 0.0011$ (mutant arm on single substitutions). Structure tokens therefore carry genuine mutation-relevant information; the contribution is small under random splitting and becomes large once position memorization is removed from the evaluation, consistent with the position channel, available to both input modes, saturating most of the random-split task.

Table 4: Real-3Di versus structure-masked SaProt inputs (paired per-dataset differences at layer 32, combined features; Wilcoxon signed-rank p). The benefit of structure tokens is small under the random split (significant for the masked_wt and mutant arms) and large and highly significant under all leakage-free splits. Mutant-arm entries under the leakage-free splits (†) are computed on single-substitution variants.

	Random		**Leakage-free**		
Arm	$\Delta\rho$	p	Modulo	Contiguous	Pos.-grouped
masked_wt	+0.003	2.2×10^{-2}	+0.073 (3.5×10^{-7})	+0.120 (8.7×10^{-8})	+0.071 (8.4×10^{-7})
unmasked_wt	+0.002	9.1×10^{-1}	+0.060 (4.6×10^{-6})	+0.086 (2.1×10^{-6})	+0.059 (1.2×10^{-5})
mutant	+0.032	2.5×10^{-6}	+0.057 (1.0×10^{-3})†	+0.132 (4.8×10^{-7})†	+0.058 (1.1×10^{-3})†

4 Discussion

The central result of this study has two parts. Under the standard random-split protocol, the 33 transformer layers of SaProt and ESM-2 show near-flat mutation effect prediction performance, but the position-lookup control shows that this flatness is largely position memorization, a property of the evaluation rather than of the representations. Under leakage-free evaluation, performance rises substantially with depth, the best layers are the last few (layers 27–32), the final layer is within $\Delta\rho = 0.019$ of the best layer (point estimates; 90% upper bounds up to 0.036), and structure tokens deliver a measurable benefit (+0.06 to +0.13). Multi-scale pooling did not reliably improve on the final layer under leakage-free evaluation (1 of 96 comparisons significantly better). The practical conclusion is therefore not that multi-scale embeddings are never useful, but that for mutation effect prediction with these 650M models depth matters and the final layer is a sufficient choice.

4.1 Why Does Performance Increase With Depth?

We propose that the depth trend reflects the MLM pre-training objective. All layers receive gradients through the same cross-entropy loss computed at the final LM head, so successive layers are optimized to move representations toward the conditional amino-acid distributions that the LM head predicts—the quantities that a mutation-effect readout ultimately uses. Mutation-relevant signal therefore accumulates with depth rather than being distributed across specialized layers, and the last layers carry most of it. Under random-split evaluation this trend is hidden because position memorization dominates the task.

The cosine similarity between layers decays monotonically with distance and, for SaProt, collapses to approximately zero only between the most distant layers, while the linear CKA between the same layers remains 0.78. This combination, a near-rotation that preserves linearly decodable information, is consistent with the effect of residual connections [19]: because residual streams accumulate representations from all preceding layers, late layers retain a linearly accessible trace of early-layer content even as their raw coordinates rotate.

This interpretation is consistent with the logit lens hypothesis [10, 25], which posits that intermediate layers in transformer language models progressively refine token predictions toward the final output distribution, rather than constructing qualitatively different representations. The lens curves support this directly for both models: zero-shot scoring improves with depth

and peaks at the final layer. The near-orthogonality of the most distant SaProt layers ($\cos \approx -0.01$) alongside their high linear CKA (0.78) suggests that linearly decodable information is retained while representations move through successive coordinate frames, even as mutation-relevant signal continues to accumulate.

In NLP transformers, by contrast, probing classifiers recover different linguistic features at different depths [9, 8]. The comparison is limited: we probe a single prediction target, which would be expected to give one peaked curve, whereas NLP studies probe many distinct features, and early PLM layers are known to encode local physicochemical and secondary-structure properties [12]. Whether PLM layers specialize less than NLP layers is therefore a hypothesis that would require probing several distinct protein properties under leakage-free splits.

4.1.1 What Separates SaProt from Standard ESM-2?

Under a single fixed protocol the two models differ mainly in two measured ways: how much their final representations retain early-layer geometry (ESM-2 distant-layer cosine 0.54 with CKA 0.41; SaProt cosine -0.01 with CKA 0.78), and the benefit conferred by structure tokens. The directly measured difference is the structure-token benefit: real 3Di inputs improve leakage-free probing by $+0.06$ to $+0.13$ while changing random-split performance only slightly. Structure information is genuinely present and genuinely useful, but leakage-free evaluation is needed to measure it.

This framing carries a testable prediction: fine-tuning standard ESM-2 with structural objectives (e.g., contact prediction or fold classification) should reproduce a similar leakage-free advantage over the sequence-only baseline. This remains a mechanistic hypothesis rather than a direct result of the present experiments.

4.2 Reconciling with Prior Layer-Selection Studies

Our findings do not contradict the observation that different layers may be optimal for different *task types*. Kumar and Jha [11] demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, achieving 32% higher clustering quality and superior classification accuracy. Our leakage-free results are broadly in line with this observation: for mutation effect prediction the best layers lie in layers 27–32 and the late layers are close to the final layer, consistent with the finding that layers 20–33 perform well. The difference is one of degree: for kinase function prediction, intermediate layers outperformed the final layer, whereas for mutation effect prediction the final layer is within a small margin of the best layer. The flat random-split profile is a property of the evaluation and is not informative for such comparisons.

Similarly, the 12% Spearman improvement reported by Alsamkary, Elshaffei et al. [13] for attention-based fusion architectures was observed in the context of multi-chain protein–protein interaction prediction, a task that differs substantially from per-residue mutation effect scoring. The benefit of sophisticated pooling may therefore be task-dependent rather than inherent to multi-scale representation fusion. For mutation effect prediction, our results suggest that such improvements should be interpreted in light of model type, readout choice, and split protocol.

4.3 Practical Implications

For researchers using protein language model embeddings for mutation effect prediction, our results support a single default for both model types: extract the final layer. Multi-scale fusion did not reliably improve on the final layer for either model under leakage-free evaluation (one significant gain in 96 comparisons), and 1280 dimensions sufficed for both linear and nonlinear readouts; this reduces storage and computational costs by a factor of 5–10 compared with concatenating 6–8 layers (up to 10,240 dimensions). Where additional effort is available, it is better spent on the readout than on fusion, at least for the sequence-only model: for ESM-2, last-layer LightGBM exceeded last-layer Ridge by +0.045 (paired, $p = 0.011$) and reached $\rho = 0.508$, the best pooling configuration overall, whereas for SaProt the same contrast was not significant; this ranking holds under the modulo split and reverses under the random split.

Table 5 quantifies the practical trade-offs. Concat 5L uses 4.9× more features than single-layer approaches and its Ridge covariance matrix has 24× more entries ($1{,}322^2$ versus $6{,}442^2$). We did not time readout fitting separately from embedding extraction: the end-to-end time per dataset is dominated by extraction, which requires one forward pass with all 33 layers collected for every strategy, and is therefore similar across strategies. Concat 5L offered no significant benefit on either model in our mutant-arm benchmark and was matched or exceeded on ESM-2 by switching from Ridge to LightGBM on a single layer. Storing Concat 5L embeddings for a typical DMS benchmark (∼10,000 variants) requires 258 MB versus 53 MB for a single layer, a gap that widens linearly with dataset size and becomes substantial for genome-scale studies.

Table 5: Computational cost comparison across pooling strategies. "Per-mutation" storage assumes float32 precision. Time estimates are per-dataset averages on an NVIDIA RTX 5090, including embedding extraction and readout training across 20 ProteinGym DMS subsets (end-to-end; extraction dominates and is identical across strategies). Best achievable ρ from Table 3 (mutant arm, modulo split).

	Last Layer	Mean L20–33	Attention	Concat 5L
Feature dimension	1,322	1,322	1,322	6,442
Storage per mutation (KB)	5.2	5.2	5.2	25.2
Ridge design matrix (5,000 vars)	6.6×10^6	same	same	3.2×10^7
Ridge covariance matrix entries	1.7×10^6	same	same	4.1×10^7
Avg. time per dataset (min)	∼3	∼3	∼3	∼3
Best achievable ρ (ESM-2 / SaProt):				
With Ridge	0.463 / 0.432	0.473 / 0.433	0.439 / 0.429	0.481 / 0.434
With LightGBM	**0.508** / **0.448**	0.457 / 0.439	0.463 / 0.433	0.495 / 0.436

Beyond the choice of layer, computational resources may be better directed toward feature types that are plausibly complementary to PLM embeddings:

1. Structural features (e.g., AlphaFold-predicted pLDDT, PAE [20], or Foldseek 3Di states [17]), which complement sequence-based embeddings;

2. Sequence-level representations that capture global protein properties beyond per-residue features;

3. Multiple sequence alignment (MSA)-based approaches [21, 22] that leverage evolutionary information beyond what PLMs encode.

As discussed under Boundary Conditions (Section 3.5), the position-lookup control shows that standard evaluation protocols can conflate mutation-effect learning with position-level memorization; random-split results alone therefore overstate performance and conceal the depth trend, whereas ProteinGym's position-based splits expose both.

4.4 Limitations

Several limitations should be noted. First, we tested the 650M-parameter ESM-2 variant and its structure-aware derivative SaProt. Smaller (35M, 150M) and larger (3B, 15B) variants may show different degrees of layer specialization, and this remains to be tested. Second, our experiments are restricted to per-residue mutation effect prediction; for other tasks such as kinase function prediction [11], subcellular localization, or protein–protein interaction scoring [13], layer-wise differences and the benefits of multi-scale fusion may be more pronounced. Third, our pooling benchmark used a 20-dataset validation subset, which is smaller than the 63-dataset probing analysis. Fourth, the SaProt pooling runs use structure-masked inputs, whereas the probing and logit lens runs include real 3Di inputs, and one dataset (influenza nucleoprotein) lacks an AFDB structure. Fifth, our feature ablation (embedding-only vs. auxiliary-only) indicates that learned embeddings dominate predictive performance in this setting, but it does not rule out the possibility that nonlinear combinations of auxiliary features could provide stronger baselines. Sixth, the maximum sample size of 10,000 variants per dataset (subsampled to 5,000 rows for regression; 11 of 63 datasets were variant-capped and 26 reached the row ceiling) may limit sensitivity to subtle layer-wise effects on the largest DMS screens (>200,000 mutations). Seventh, our linear probing approach can only detect information that is linearly separable; nonlinear interactions between layers could theoretically exist but would require a different experimental design to identify. Eighth, position-level data splitting (Section 3.5) shows that random mutation splits permit position-level leakage, which the lookup control quantifies and which motivates our reporting of leakage-free splits throughout. Ninth, under the position-based splits the mutant arm is analyzed on single substitutions, which excludes most variants of the ten datasets that contain multi-site variants; the wild-type arms are reported on the full data, in which rows of a multi-site variant share labels across folds, although their single-substitution results are similar (Supplementary Table S8). Finally, our equivalence margin ($\Delta\rho = 0.01$) was fixed before the equivalence analysis as a practical convention rather than a task-derived threshold, and because the best layer is selected on the same data, the equivalence verdicts are conservative; three of the seven equivalence verdicts nevertheless lie at the margin and depend on the interval method.

4.5 Conclusion

For mutation effect prediction with 650M protein language models, depth matters: under leakage-free evaluation, predictive performance rises with depth and the best layers are the last few. The final layer is a sufficient choice—within $\Delta\rho = 0.019$ of the best layer (point estimates; 90% upper bounds up to 0.036), a small deficit relative to the variation across datasets—and multi-scale

pooling did not reliably improve on it under leakage-free evaluation. Under the standard random split, by contrast, near-flat layer profiles are reproduced by a position lookup table with no sequence information, so random-split layer interchangeability is a property of the evaluation, not of the representations. Structure tokens add a benefit that is small under random splitting and large under leakage-free evaluation. These findings support extracting the final layer, evaluating with a position-aware split, and treating claims of layer-specific representations in PLMs with the same scrutiny that has become standard in the NLP community [8].

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Competing Interests

The authors declare no competing interests.

Data Availability

Code and results are available at `https://github.com/schwarzeteeguo-creator/esm2-layer-invariance`. An archived snapshot of the code, per-dataset results, and extracted structure tokens for this revision is deposited at Zenodo (concept DOI 10.5281/zenodo.22844290; archived version v3.1.1, DOI 10.5281/zenodo.23878920). Raw ProteinGym data: `https://github.com/OATML-Markslab/ProteinGym`.

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